Watching Evolution Happen: Digital Organisms Under Selection

Avida-ED · Michigan State University

Year 10	~30 minutes	Science · Biology

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Simulation: https://avida-ed.msu.edu/
Curriculum links: AC9S10U02 · evolution · natural selection

Learning intentions

- I can set up a population of digital organisms and apply a selective pressure.
- I can record how a trait changes across generations and identify this as evidence of natural selection.
- I can explain why some digital organisms are more likely to survive and reproduce than others under a given condition.

Getting started

Open Avida-ED and set up a fresh population of digital organisms. Choose one selective pressure to test, for example rewarding organisms that can perform a particular task, and note your starting population's overall characteristics before you run any generations.

Tasks

1. Before running any generations, describe your starting population. Do the organisms mostly look the same, or is there already variation between them?

2. Run the population forward several generations under your chosen selective pressure. What do you notice about the population's overall traits after this time, compared with the start?

3. Pause the run at several points and estimate the proportion of the population showing the favoured trait.

Generation checkpoint	Approx. proportion showing the favoured trait

4. Explain why the proportion of organisms with the favoured trait changed the way it did. Use the words 'mutation', 'selection' and 'reproduce' in your answer.

5. Reset the population and run it again with no selective pressure applied, or a different one. Does the population change in the same way? What does this tell you about the role of the selective pressure?

6. A classmate says 'the organisms are trying to evolve the useful trait.' Using evidence from your Avida-ED run, explain why this statement is misleading and how natural selection actually works.

7. Identify one part of your Avida-ED results that provides direct evidence for evolution by natural selection, rather than just an assumption.

Extension (optional)

Real bacteria evolving antibiotic resistance follow a similar pattern to what you saw in Avida-ED. Explain the similarities between your simulated population and a real population of bacteria exposed to an antibiotic.